

FYS5419/9419 March 11



FYS5419/9419 March 11

$$N = 2$$

$$\text{---} \times \times \text{---} + \epsilon \quad \uparrow$$

$$\text{---} \bigcirc \bigcirc \text{---} - \epsilon \quad \downarrow$$

$\varphi \quad 1 \quad 2$

$$J = 1 \quad J_z = -1, 0, 1$$

$$|J=1, J_z\rangle : \{ |1-1\rangle, |10\rangle, |11\rangle \}$$

$$\# \text{ configs} = \binom{4}{2} = 6$$

$$\langle JJ_z | H | JJ_z' \rangle$$

$$|J=1-1\rangle \text{ and } |J=1+1\rangle$$

$$\begin{bmatrix} -\epsilon & V & 0 \\ V & \epsilon & 0 \\ 0 & 0 & \bar{W} \end{bmatrix} \begin{matrix} \leftarrow |J=1J_z=-1\rangle \\ |J=1J_z=+1\rangle \\ |J=1J_z=0\rangle \end{matrix}$$

$$\begin{bmatrix} -\epsilon & V \\ V & \epsilon \end{bmatrix} \rightarrow \text{one qubit problem}$$

$|0\rangle, |1\rangle$

3x3 Hamiltonian, 9 unit

basis $|00\rangle$ $|01\rangle$ $|10\rangle$ $|11\rangle$

$2^2 = 4$ comp basis states

$J=2$ P $\begin{array}{cccc} 1 & 2 & 3 & 4 \\ \hline \times & \times & \times & \times \end{array}$

$\begin{array}{cccc} \hline 0 & 0 & 0 & 0 \end{array} - 2 \checkmark$
 P $\begin{array}{cccc} 1 & 2 & 3 & 4 \end{array}$

$J=2$ $J_z = -2, -1, 0, 1, 2$

config $= \binom{8}{4} = 70$

$$\langle j_z = 2 | \kappa | j_z = 2 \rangle \neq 0$$

$$\langle J_z = 2 | \sigma | J_z = 0 \rangle \neq 0$$

$$\underline{\langle J_z = 2 | \psi | J_z = -2 \rangle} \neq 0$$

and similarly for $\nabla z = 0$
 $\nabla z = -2$

$$\langle J_2 = \pm 1 | H | J_2 = \text{even} \rangle = 0$$

$$\left[\begin{array}{ccc|cc}
 \hline -2\varepsilon & \sqrt{6}V & 0 & & \\
 \hline \sqrt{6}V & 4W & \sqrt{6}V & & \\
 \hline 0 & \sqrt{6}V & 2\varepsilon & & \\
 \hline & & & -\varepsilon+W & 3V \\
 & & & 3V & \varepsilon+3W \\
 \hline
 \end{array} \right]$$

$|00\rangle$ $|01\rangle$ $|10\rangle$
 $|11\rangle$ 2 qubit case

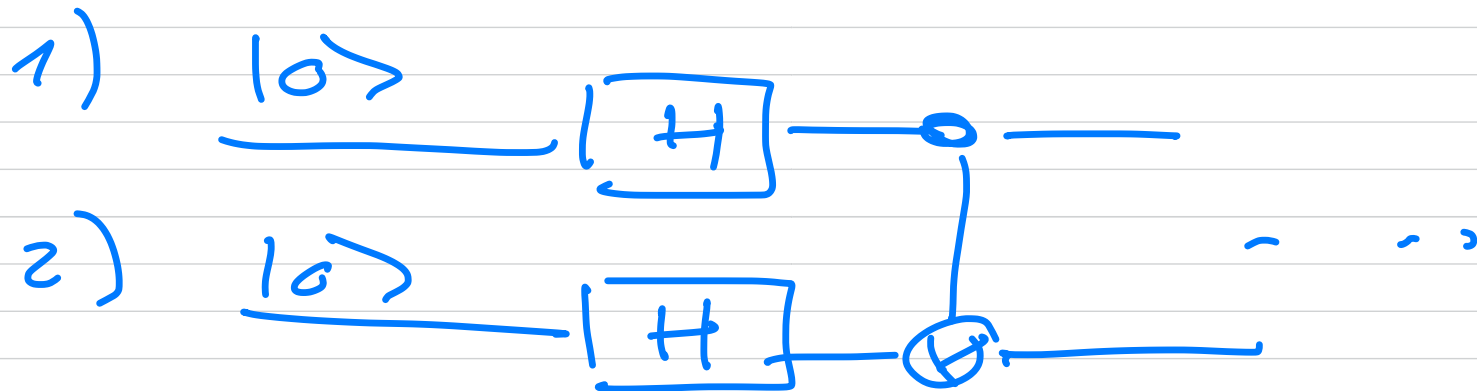
$\sqrt{J_z = \pm 1}$
 one qubit $|0\rangle, |1\rangle$

Mapping 4×4 H-matrix
 we get Pauli strings of
 the type

$$Z_1 \otimes I_2, \quad I_1 \otimes Z_2, \quad I_1 \otimes I_2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad Z_1 \otimes Z_2 \dots X_1 \otimes X_2$$

$$\dots \quad Y_1 \otimes Y_2$$



Measurement

$$P = U^\dagger M U$$

one-qubit case example

$$P = X = H Z H$$

$$S = H S^\dagger Z H S$$

$$Z \otimes \mathbb{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$$(Z \otimes I) |00\rangle = + |00\rangle \quad \text{qubit 1}$$

$$(Z \otimes I) |01\rangle = + |01\rangle$$

$$(Z \otimes I) |10\rangle = - |10\rangle$$

$$(Z \otimes I) |11\rangle = - |11\rangle$$

$$(I \otimes Z) |00\rangle = |00\rangle$$

$$(I \otimes Z) |01\rangle = -|01\rangle$$

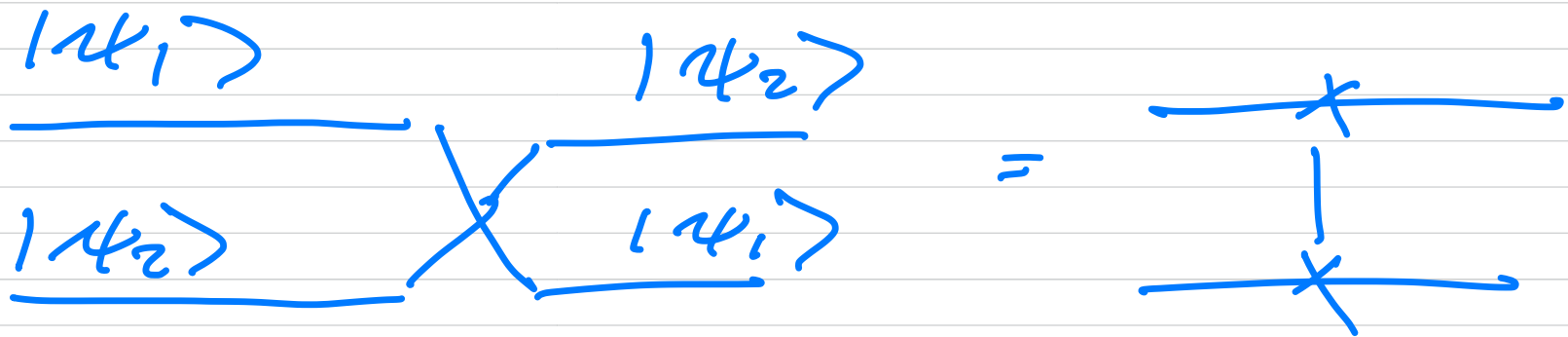
$$(I \otimes Z) |10\rangle = |10\rangle$$

$$(I \otimes Z) |11\rangle = -|11\rangle$$

$$(I \otimes Z) = U^\dagger (Z \otimes I) U$$

$$\text{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \left((I \otimes I) + (X \otimes X) + (Y \otimes Y) + (Z \otimes Z) \right) / 2$$



$$\text{SWAP}|00\rangle = |00\rangle$$

$$\text{SWAP}|11\rangle = |11\rangle$$

$$\text{SWAP}|01\rangle = |10\rangle$$

$$\text{SWAP}|10\rangle = |01\rangle$$

$$I \otimes Z = \text{SWAP} (Z \otimes I) \text{SWAP}$$

$$Z \otimes Z \rightsquigarrow U^\dagger M U$$

$\uparrow$
 $Z \otimes I$

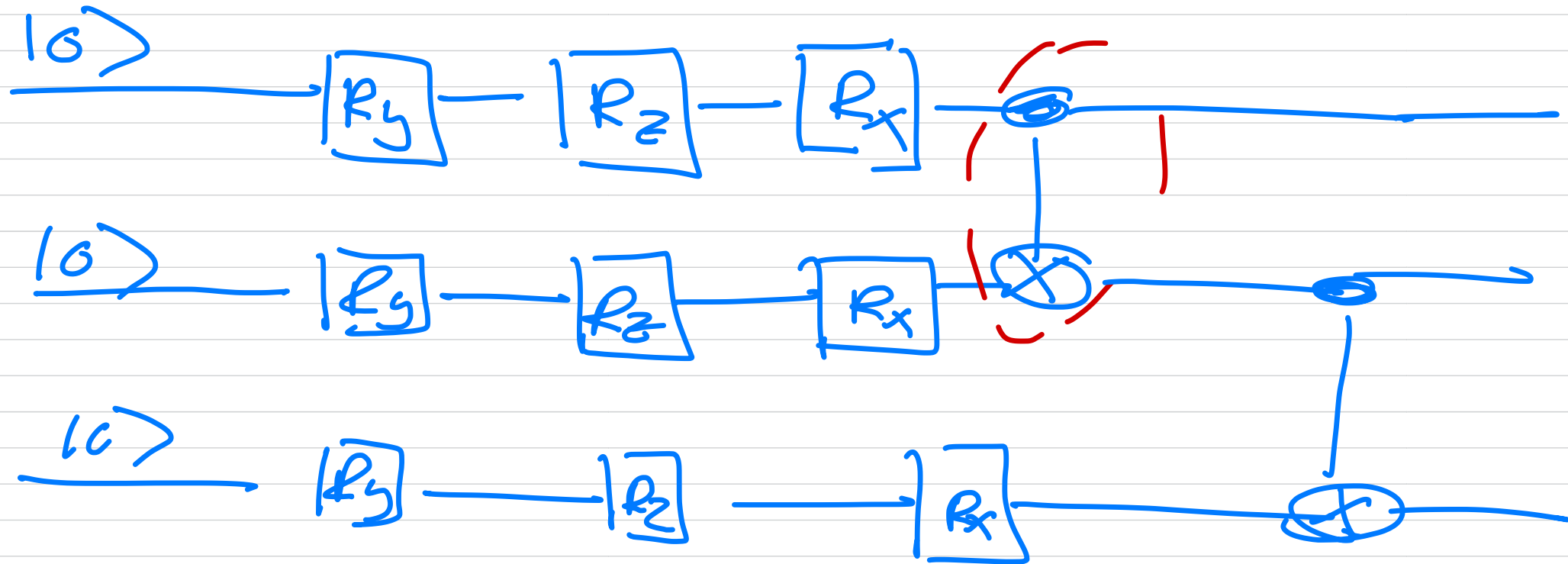
$$U = CX_{10}$$

$$CX_{10} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$CX_{c1} = C N C^T = \begin{bmatrix} 1 & 0 \\ 0 & x \end{bmatrix}$$

3 qubits

entangle



Parameterize a quantum circuit

$$f'(x_{k+1}) \approx \frac{f(x_{k+1}) - f(x_k)}{x_{k+1} - x_k}$$

$$f(x_5) = 0$$

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$\nabla f(x_5) = 0$$

Jordan-Wigner transf

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_I$$

↙

$$\sum_i \varepsilon_i a_i^\dagger a_i + \frac{1}{4} \sum_{pqrs} \langle pq | \mathcal{H} | rs \rangle_{AS}$$

$$\times a_p^\dagger a_q^\dagger a_s a_r$$

$$\{a_p, a_q^\dagger\} = a_p a_q^\dagger + a_q^\dagger a_p = \delta_{pq}$$

$$\{a_p, a_q\} = \{a_p^\dagger, a_q^\dagger\} = 0$$

$$a_p^\dagger |0\rangle = |p\rangle \quad a_p |p\rangle = |0\rangle$$

$$a_p |0\rangle = 0 \quad a_p^\dagger |p\rangle = 0$$

$$a_1^\dagger a_2^\dagger |0\rangle = |12\rangle$$

$$\hat{N}_p = a_p^\dagger a_p$$

Pauli' strings

$$\Delta^\dagger = \frac{x - iy}{z} = |1\rangle \langle 0|$$

$$Y = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$|1\rangle\langle 0| = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$|1\rangle\langle \underbrace{0|0}_1\rangle$$

$$\sigma^+ |0\rangle = |1\rangle$$

$$\sigma^+ \sim a_p^\dagger$$

$$\sigma^- = \frac{X + iY}{2} = |0\rangle\langle 1|$$

$$\sigma^- |1\rangle = |0\rangle \langle 1|1\rangle = |0\rangle$$

$$\sigma^- \sim a_p$$

$$\sigma^- |0\rangle = |0\rangle \langle 1|0\rangle = 0$$